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## Information processing apparatus, non-transitory computer readable medium storing information processing program, and information processing method


### Abstract

An information processing apparatus includes: a processor configured to: in a case where a first page to which a first code including process information, which represents information for processing a document and in which at least another page other than the first page is set as a process target, is added and a second page to which a second code including process information, which represents information for processing a document and in which the second page is set as the process target, is added are captured, execute a process represented by the process information of the first code without executing a process represented by the process information of the second code in a case where the process represented by the process information of the first code conflicts with the process represented by the process information of the second code.



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## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 5506697 | 12/1995 | Li | 358/448 | H04N 1/00363 |
| 2017/0277926 | 12/2016 | Yoshizaki | N/A | H04N 1/00334 |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 3962759 | 12/2006 | JP | N/A |

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-012158 filed Jan. 28, 2022.
BACKGROUND
(i) Technical Field
(2) The present invention relates to an information processing apparatus, a non-transitory computer readable medium storing an information processing program, and an information processing method.
(ii) Related Art
(3) JP3962759B discloses a document processing apparatus that digitizes a block of documents including at least one document group configured with one or more documents, in which a code is printed or attached to at least a document at a head in the document group, the code includes an identification number for specifying the document group and attribute information which is information on a process to be executed on the digitized document, the attribute information includes information indicating a size of the document constituting the document group and information indicating a necessity of correction for an image related to the document, the document processing apparatus includes a document scanning section that acquires data of the digitized document and scans the code attached to the document, a processing content determination section that determines contents of the process to be executed on the digitized document in the document

group based on the attribute information included in the scanned code, and an error status of the scanned document, and a document processing section that executes the determined process, and the processing content determination section includes a document group sorting unit that specifies a document group and sorts the digitized document for each document group based on the identification number, an error status determination unit that detects a size of the digitized document and determines that the document has an error in a case where the detected size and the size indicated by the attribute information are compared with each other and do not match, and a group processing determining unit that detects, in a case where the error status determination unit determines the error and the information included in the attribute information indicates that the correction is required, an inclination of the digitized document, determines, in a case where it is determined that the digitized document has no inclination, the process of correcting only the size of the document, and determines, in a case where it is determined that the digitized document has the inclination, the process of rotating and correcting the document.

SUMMARY

(4) There is known a technique for capturing a document as image data by a scanning process and executing a process on the captured image data. Here, there is a section that preliminarily adds a physical code including process information representing the process, for example, a quick response (QR) code (registered trademark) to a page included in the document. In a case where the document is captured by the scan process, the process information is extracted from the QR code (registered trademark) added to the page, and the process represented by the process information is executed.

(5) Meanwhile, the QR code (registered trademark) includes a QR code (registered trademark) (hereinafter, referred to as a “first code”) in which a process target of the process information including the QR code (registered trademark) is a set of pages including at least other pages other than the page to which the QR code (registered trademark) is added, and a QR code (registered trademark) (hereinafter, referred to as a “second code”) in which the process target of the process information including the QR code (registered trademark) is a page to which the QR code (registered trademark) is added.

(6) Here, when a page to which the first code is added is captured and a process represented by process information included in the first code is executed on the subsequent page, a page to which the second code is added is captured, and a process represented by process information included in the second code is executed on the page to which the second code is added, in some cases. In a case where the process represented by the process information included in the first code and the process information included in the second code conflict with each other, for example, in a case where contents of the process is a process of storing image data in different storage destinations, a contradiction will occur and the process cannot be normally executed.

(7) Aspects of non-limiting embodiments of the present disclosure relate to an information processing apparatus, a non-transitory computer readable medium storing an information processing program, and an information processing method that normally execute, even in a case where a page to which a first code is added and a page to which a second code is added are mixed and a process represented by process information included in the first code and a process represented by process information included in the second code conflicts with other, the processes.

(8) Aspects of certain non-limiting embodiments of the present disclosure overcome the above disadvantages and/or other disadvantages not described above. However, aspects of the non-limiting embodiments are not required to overcome the disadvantages described above, and aspects of the non-limiting embodiments of the present disclosure may not overcome any of the disadvantages described above.

(9) According to an aspect of the present disclosure, there is provided an information processing apparatus including: a processor configured to: in a case where a first page to which a first code including process information, which represents information for processing a document and in

which at least another page other than the first page is set as a process target, is added and a second page to which a second code including process information, which represents information for processing a document and in which the second page is set as the process target, is added are captured, execute a process represented by the process information of the first code without executing a process represented by the process information of the second code in a case where the process represented by the process information of the first code conflicts with the process represented by the process information of the second code.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:

(2) FIG. **1** is a schematic diagram illustrating an example of a configuration of an information processing system according to each exemplary embodiment;

(3) FIG. **2** is a schematic diagram illustrating an example of a capturing process on a document by using a code added to a face sheet according to each exemplary embodiment;

(4) FIG. **3** is a schematic diagram illustrating an example of a capturing process on a document by using a code individually added to each page according to each exemplary embodiment;

(5) FIG. **4** is a block diagram illustrating an example of a hardware configuration of an information processing apparatus according to each exemplary embodiment;

(6) FIG. **5** is a block diagram illustrating an example of a functional configuration of an information processing apparatus according to a first exemplary embodiment;

(7) FIG. **6** is a schematic diagram illustrating an example of a process of capturing a plurality of documents according to each exemplary embodiment;

(8) FIG. **7** is a schematic diagram illustrating another example of the process of capturing the plurality of documents according to each exemplary embodiment;

(9) FIG. **8** is a schematic diagram illustrating still another example of the process of capturing the plurality of documents according to each exemplary embodiment;

(10) FIG. **9** is a schematic diagram illustrating still another example of the process of capturing the plurality of documents according to each exemplary embodiment;

(11) FIG. **10** is a schematic diagram illustrating still another example of the process of capturing the plurality of documents according to each exemplary embodiment;

(12) FIG. **11** is a flowchart illustrating an example of a document capturing process according to the first exemplary embodiment;

(13) FIG. **12** is a flowchart illustrating an example of a process of setting a target and a storage destination according to the first exemplary embodiment;

(14) FIG. **13** is a block diagram illustrating an example of a functional configuration of an information processing apparatus according to a second exemplary embodiment;

(15) FIG. **14** is a schematic diagram illustrating an example of a captured document display screen according to the second exemplary embodiment;

(16) FIG. **15** is a flowchart illustrating an example of a capturing process on a document according to the second exemplary embodiment; and

(17) FIG. **16** is a schematic diagram illustrating an example of a captured document display screen according to a modification example of the second exemplary embodiment.

DETAILED DESCRIPTION

First Exemplary Embodiment

(18) Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings. FIG. **1** is a schematic diagram illustrating an example of a

configuration of an information processing system **1** according to the present exemplary embodiment.
(19) As an example, as illustrated in FIG. **1**, the information processing system **1** includes a multi-function device **2** having a scanner function and an information processing apparatus **10** which is a server. The multi-function device **2** and the information processing apparatus **10** are connected to each other via a network N. In the present exemplary embodiment, a mode in which the information processing apparatus **10** is a server will be described. Meanwhile, the exemplary embodiment is not limited thereto. The information processing apparatus **10** may be mounted on the multi-function device **2**, or may be a terminal such as a personal computer.
(20) The multi-function device **2** scans a document including one or a plurality of pages according to an instruction of a user, captures the document as image data, and transmits the image data to the information processing apparatus **10**.
(21) The information processing apparatus **10** acquires the image data from the multi-function device **2**, and scans a quick response (QR) code (registered trademark) (hereinafter, referred to as a "code") added to the page included in the document from the acquired image data. According to information on a process to be performed on the document included in the code (hereinafter, referred to as "process information"), the information processing apparatus **10** has a function of executing the process. In the present exemplary embodiment, a mode in which a QR code (registered trademark) is added as the code will be described. Meanwhile, the exemplary embodiment is not limited thereto. The code may be a barcode (one-dimensional code), text information, a figure, or other information, as long as the process information can be recognized by the information processing apparatus **10**.
(22) The process information according to the present exemplary embodiment includes a storage destination for storing image data and a type of code (hereinafter, referred to as a "code type") indicating a process of capturing a target to be stored in the storage destination. The code type includes a "face sheet process" representing a process of capturing a set of pages using a face sheet, and an "individual process" representing a process of capturing a document individually for each page. In the following, a code indicating the face sheet process is referred to as a "face sheet code", and a code indicating the individual process is referred to as an "individual code". Here, the face sheet code is an example of a first code, and the individual code is an example of a second code.
(23) In a case where a code type included in a code is a face sheet process, the information processing apparatus **10** executes the face sheet process of setting image data related to a page after a page from which the code is scanned as a target, and storing the image data of the set target in a storage destination included in process information. Further, in a case where the code type is an individual process, the information processing apparatus **10** executes the individual process of setting the image data related to the page from which the code is scanned as the target, and storing the image data of the set target in the storage destination included in the process information.
(24) FIG. **2** is a schematic diagram illustrating an example of a face sheet process according to the present exemplary embodiment. The face sheet process is a process of storing, from a document including a face sheet **31** to which a face sheet code **30** is added at 1-page and pages **32** of 2-page and the subsequent pages as targets, image data **33** corresponding to the page **32** according to process information. Here, the face sheet **31** is an example of a first page.
(25) Further, FIG. **3** is a schematic diagram illustrating an example of an individual process according to the present exemplary embodiment. The individual process is a process of storing, from a document including each page **35** to which an individual code **34** indicating the individual process is added, each image data **36** corresponding to the page **35** according to process information. Here, each page **35** to which the individual code **34** is added is an example of a second page.
(26) In the present exemplary embodiment, a mode in which a plurality of documents including pages to which each of the face sheet code **30** and the individual code **34** is added are captured and

corresponding image data is stored will be described.
(27) Next, a hardware configuration of the information processing apparatus **10** will be described with reference to FIG. **4**. FIG. **4** is a block diagram illustrating an example of the hardware configuration of the information processing apparatus **10** according to the present exemplary embodiment.
(28) As illustrated in FIG. **4**, the information processing apparatus **10** according to the present exemplary embodiment is configured to include a central processing unit (CPU) **11**, a read only memory (ROM) **12**, a random access memory (RAM) **13**, a storage **14**, an input unit **15**, a monitor **16**, and a communication interface (communication I/F) **17**. The CPU **11**, the ROM **12**, the RAM **13**, the storage **14**, the input unit **15**, the monitor **16**, and the communication I/F **17** are connected to each other via a bus **19**. Here, the CPU **11** is an example of a processor.
(29) The CPU **11** collectively controls the entire information processing apparatus **10**. The ROM **12** stores various programs including an information processing program, data, and the like used in the present exemplary embodiment. The RAM **13** is a memory used as a work area when executing various programs. The CPU **11** expands the program stored in the ROM **12** into the RAM **13** to execute the process of capturing the document.
(30) The storage **14** is a hard disk drive (HDD), a solid state drive (SSD), a flash memory, or the like as an example. The input unit **15** is a touch panel, a keyboard, or the like that accepts a text input and a target selection. The monitor **16** displays texts and images. The communication I/F **17** transmits and receives data.
(31) Next, a functional configuration of the information processing apparatus **10** will be described with reference to FIG. **5**. FIG. **5** is a block diagram illustrating an example of a functional configuration of the information processing apparatus **10** according to the present exemplary embodiment.
(32) As an example, as illustrated in FIG. **5**, the information processing apparatus **10** includes an acquisition unit **21**, an extraction unit **22**, a setting unit **23**, and a processing unit **24**A. In a case where the CPU **11** executes the information processing program to function as the acquisition unit **21**, the extraction unit **22**, the setting unit **23**, and the processing unit **24**A.
(33) The acquisition unit **21** acquires image data from the multi-function device **2**. Here, the acquired image data is image data obtained by capturing a plurality of documents including pages to which each of the face sheet code **30** representing the face sheet process and the individual code **34** representing the individual process is added.
(34) The extraction unit **22** scans a code included in the image data, and extracts a code type set in the code and a storage destination of the image data, as process information.
(35) The setting unit **23** sets the extracted process information in the information processing apparatus **10**. Specifically, the setting unit **23** sets a target of the image data according to the extracted storage destination and the extracted code type. For example, in a case where the code type is the face sheet process, the setting unit **23** sets the image data corresponding to a page after a page from which the code type is extracted as the target, and in a case where the code type is the individual process, the setting unit **23** sets the image data corresponding to the page from which the code is extracted as the target.
(36) Further, in a case where process information on the face sheet process is already set as the setting of the information processing apparatus **10**, the setting unit **23** does not set process information on the individual process as the setting of the information processing apparatus **10** even in a case where the code type indicating the individual process is extracted.
(37) The processing unit **24**A executes a process of storing the target image data in the set storage destination.
(38) Next, before an action of the information processing apparatus **10** according to the present exemplary embodiment is described, a method of processing a document including a plurality of codes will be described with reference to FIGS. **6** to **10**. FIGS. **6** to **10** are schematic diagrams

illustrating an example of a process of capturing a plurality of documents according to the present exemplary embodiment.
(39) The information processing apparatus **10** according to the present exemplary embodiment scans a code added to a page included in the document, and extracts a code type and a storage destination as process information. FIGS. **6** to **8** illustrate a mode for extracting a face sheet code **30** in which a folder A is set in a storage destination as process information, and individual codes **34**A and **34**B in which a folder B is set in a storage destination.
(40) First, with reference to FIG. **6**, a mode in which the documents are captured in an order of a page **35**A to which the individual code **34**A is added, a page **35**B to which the individual code **34**B is added, and a face sheet **31** to which the face sheet code **30** is added, as a capturing order will be described.
(41) The information processing apparatus **10** extracts the individual code **34**A from the page **35**A, and stores image data **36**A corresponding to the page **35**A in the folder B, according to process information included in the individual code **34**A. In the same manner, the information processing apparatus **10** extracts the individual code **34**B from the page **35**B, and stores image data **36**B corresponding to the page **35**B in the folder B, according to process information included in the individual code **34**B. Further, in a case where the face sheet code **30** is extracted, the information processing apparatus **10** stores the image data **33** corresponding to the page **32** after the face sheet **31** in the folder A, according to process information included in the face sheet code **30**.
(42) Next, with reference to FIG. **7**, a mode in which the documents are captured in an order of the page **35**A to which the individual code **34**A is added, the face sheet **31** to which the face sheet code **30** is added, and the page **35**B to which the individual code **34**B is added, as the capturing order will be described.
(43) The information processing apparatus **10** extracts the individual code **34**A from the page **35**A, and stores image data **36**A corresponding to the page **35**A in the folder B, according to the process information included in the individual code **34**A. In a case where the face sheet code **30** is extracted, the information processing apparatus **10** stores the image data **33** corresponding to the page **32** after the face sheet **31** in the folder A, according to the process information included in the face sheet code **30**. Further, in a case where the individual code **34**B is extracted after the face sheet code **30** is extracted, the information processing apparatus **10** does not set process information on the individual code **34**B, and stores the image data **36**B corresponding to the page **35**B in the folder A already set in the information processing apparatus **10**.
(44) Next, with reference to FIG. **8**, a mode in which the documents are captured in an order of the face sheet **31** to which the face sheet code **30** is added, the page **35**A to which the individual code **34**A is added, and the page **35**B to which the individual code **34**B is added, as the capturing order will be described.
(45) In a case where the face sheet code **30** is extracted, the information processing apparatus **10** stores the image data **33** corresponding to the page **32** after the face sheet **31** in the folder A, according to the process information included in the face sheet code **30**. In a case where the individual code **34**A and the individual code **34**B are extracted after the face sheet code **30** is extracted, the information processing apparatus **10** does not set process information on the individual code **34**A and the individual code **34**B. The information processing apparatus **10** respectively stores the image data **36**A and the image data **36**B corresponding to the page **35**A and the page **35**B in the folder A, according to process information on the face sheet code **30** already set in the information processing apparatus **10**.
(46) As illustrated in FIGS. **6** to **8**, the information processing apparatus **10** sets the target and the storage destination according to the individual code by using the process information included in the extracted code until the face sheet code **30** is extracted. Then, the information processing apparatus **10** stores the target image data in the set storage destination.
(47) For example, in a case where the individual code **34** is extracted before the face sheet code **30**

is extracted, the information processing apparatus **10** sets the storage destination and the target included in the individual code **34** as the process information, and stores the image data **36** corresponding to the page **35**, which is the target, in the set storage destination. In a case where the face sheet code **30** is extracted, the information processing apparatus **10** sets the storage destination and the target included in the face sheet code **30** as the process information, and stores the image data **33** corresponding to the page **32**, which is the target, in the set storage destination. Even in a case where the individual code **34** is extracted after the face sheet code **30** is extracted, the information processing apparatus **10** does not set process information included in the individual code **34**, and applies the setting related to the face sheet code **30** already set to perform the process.
(48) Next, with reference to FIG. **9**, a mode in which the documents are captured in an order of a face sheet **31**A to which a face sheet code **30**A is added and a face sheet **31**B to which a face sheet code **30**B is added, as the capturing order will be described. In the following, a mode for extracting the face sheet code **30**A in which the folder A is set as a storage destination and the face sheet code **30**B in which the folder B is set as the storage destination, as process information will be described.
(49) The information processing apparatus **10** extracts the face sheet code **30**A from the face sheet **31**A, and stores the image data **33**A corresponding to the page **32**A in the folder A, according to process information included in the face sheet code **30**A. The information processing apparatus **10** extracts the face sheet code **30**A, then extracts the face sheet code **30**B, and stores the image data **33**B corresponding to a page **32**B in the folder B, according to process information included in the face sheet code **30**B.
(50) Next, with reference to FIG. **10**, a mode in which the documents are captured in an order of the page **35**A to which the individual code **34**A is added, the face sheet **31**A to which the face sheet code **30**A is added, the page **35**B to which the individual code **34**B is added, the face sheet **31**B to which the face sheet code **30**B is added, and a page **35**C to which an individual code **34**C is added, as the capturing order will be described. In the following, a mode in which the face sheet code **30**A in which the folder A is set as a storage destination, the face sheet code **30**B in which a folder C is set as the storage destination, and the individual code **34**A, the individual code **34**B, and the individual code **34**C in which the folder B is set as the storage destination, as process information are extracted will be described.
(51) The information processing apparatus **10** extracts the individual code **34**A, and stores the image data **36**A corresponding to the page **35**A in the folder B, according to the process information.
(52) Next, in a case where the face sheet code **30**A is extracted, the information processing apparatus **10** stores the image data **33**A corresponding to the page **32**A after the face sheet **31**A in the folder A, according to the process information included in the face sheet code **30**. Further, in a case where the individual code **34**B is extracted after the face sheet code **30**A is extracted, the information processing apparatus **10** does not set process information on the individual code **34**B, and stores the image data **36**B corresponding to the page **35**B in the folder A already set in the information processing apparatus **10**.
(53) Further, in a case where the face sheet code **30**B is extracted after the face sheet code **30**A is extracted, the information processing apparatus **10** stores the image data **33**B corresponding to the page **32**B in the folder C, according to the process information included in the face sheet code **30**B. Further, in a case where the individual code **34**C is extracted after the face sheet code **30**B is extracted, the information processing apparatus **10** does not set process information on the individual code **34**C, and stores image data **36**C corresponding to the page **35**C, in the folder C already set in the information processing apparatus **10**.
(54) As illustrated in FIGS. **9** and **10**, in a case where a document includes a plurality of face sheets **31** to which the face sheet code **30** is added, the information processing apparatus **10** sets each of a storage destination and a target included in the face sheet code **30** as process information, and respectively stores the image data **33** in the set storage destination.

(55) Here, a target related to the face sheet code **30**A is from the page **32**A after the face sheet **31**A to a page before the face sheet **31**B. A target of the face sheet code **30**B is the page **32**B after the face sheet **31**B. For example, even in a case where the page **35** is included between the face sheet **31**A and the face sheet **31**B, the information processing apparatus **10** stores the image data **36** corresponding to the page **35**, according to process information included in the face sheet **31**A. Further, even in a case where the page **35**C to which the individual code **34**C is added is included after the face sheet **31**B, the information processing apparatus **10** stores the image data **36**C corresponding to the page **35**C, according to process information included in the face sheet **31**B.
(56) Next, an action of the information processing apparatus **10** according to the present exemplary embodiment will be described with reference to FIGS. **11** and **12**. FIG. **11** is a flowchart illustrating an example of a document capturing process according to the present exemplary embodiment. In a case where the CPU **11** reads an information processing program from the ROM **12** or the storage **14**, and executes the information processing program to execute a document capturing process illustrated in FIG. **11**. The document capturing process illustrated in FIG. **11** is executed, for example, in a case where an instruction to execute a process and image data are input from the multi-function device **2**.
(57) In step S**101**, the CPU **11** acquires image data from the multi-function device **2**. Here, for the acquired image data, the CPU **11** sets 1-page of a document related to the image data as a scanning page for scanning a code.
(58) In step S**102**, the CPU **11** determines whether or not the face sheet code **30** or a code indicating the individual code **34** is scanned from the scanning page. In a case where the code is scanned (YES in step S**102**), the CPU **11** proceeds to step S**103**. On the other hand, in a case where the code cannot be scanned from the scanning page (NO in step S**102**), the CPU **11** proceeds to step S**104**.
(59) In step S**103**, the CPU **11** uses process information included in the scanned code to execute a setting process of setting a target and a storage destination according to the process information. The setting process will be described in detail in FIG. **12**, which will be described below.
(60) In step S**104**, the CPU **11** uses the target and the storage destination set in the information processing apparatus **10** to store the image data corresponding to the scanning page in the storage destination.
(61) In step S**105**, the CPU **11** determines whether or not the next page exists in the document related to the acquired image data. In a case where the next page does not exist in the document related to the image data (YES in step S**105**), the CPU **11** ends the process of capturing the document. On the other hand, in a case where the next page exists in the image data (NO in step S**105**), the CPU **11** proceeds to step S**106**.
(62) In step S**106**, the CPU **11** sets the next page to be scanned as the scanning page in the document related to the acquired image data.
(63) Next, the setting process will be described with reference to FIG. **12**. FIG. **12** is a flowchart illustrating an example of a process of setting a target and a storage destination according to the present exemplary embodiment. In a case where the CPU **11** reads a setting program from the ROM **12** or the storage **14**, and executes the setting program to execute the setting process illustrated in FIG. **12**. The setting process illustrated in FIG. **12** is executed, for example, in a case where a code added to a page included in a document is scanned.
(64) In step S**201**, the CPU **11** extracts a code type and a storage destination from the code as process information.
(65) In step S**202**, the CPU **11** determines whether or not the code type is a face sheet process. In a case where the code type is the face sheet process (YES in step S**202**), the CPU **11** proceeds to step S**203**. On the other hand, in a case where the code type is not the face sheet process (code type is an individual process) (NO in step S**202**), the CPU **11** proceeds to step S**204**.
(66) In step S**203**, the CPU **11** sets the extracted target and the extracted storage destination

according to the face sheet process, as a setting of the information processing apparatus **10**.
(67) In step S**204**, the CPU **11** determines whether or not the target according to the face sheet process is already set as the setting of the information processing apparatus **10**. In a case where the target corresponding to the face sheet process is already set (YES in step S**204**), the CPU **11** ends the setting process without setting the extracted target and the extracted storage destination according to the individual process. On the other hand, in a case where the target according to the face sheet process is not set (NO in step S**204**), the CPU **11** proceeds to step S**205**.
(68) In step S**205**, the CPU **11** sets the extracted target and the extracted storage destination according to the individual process, as the setting of the information processing apparatus **10**.
(69) As described above, according to the present exemplary embodiment, in a case where a page to which a first code is added and a page to which a second code is added are mixed, processes can be executed normally even in a case where the process represented by process information included in the first code and the process represented by process information included in the second code conflict with each other. With the above exemplary embodiment, the mode in which in a case where process information on a face sheet process is already set, process information on an individual process is not set as a setting of the information processing apparatus **10** even in a case where a code type indicating the individual process is extracted. Meanwhile, the exemplary embodiment is not limited thereto. In a case where it is determined that the process information on the face sheet process and the process information on the individual process information conflict with each other after the code type indicating the individual process is extracted and the process information on the individual process is set, the setting of the process information on individual process may be invalidated. As described above, the conflict of the process information means that the contents of the processes related to a plurality of pieces of process information do not hold at the same time.
Modification Example of First Exemplary Embodiment
(70) In the present exemplary embodiment, the mode in which in a case where pieces of process information conflict with each other, such as indicating a different storage destination, the process information on a face sheet code is preferentially set is described. Meanwhile, the exemplary embodiment is not limited thereto. A priority order may be set for each process information included in a code, and the process information may be set according to the priority order.
(71) For example, in a case where a higher priority order than process information on a face sheet code is set for process information on an individual code, even in a case where the individual code is scanned after the face sheet code is scanned, the information processing apparatus **10** sets the process information on the individual code to which the high priority is set to perform a process.
(72) Further, in the present exemplary embodiment, the mode in which a code type and a storage destination are included as the process information is described. Meanwhile, the exemplary embodiment is not limited thereto. Other settings may be included. For example, other settings may include a setting such as an optical character recognition (OCR) process, a setting of attributes given to a document, and a folding setting for designating how to fold the document. In a case where the process information on the face sheet code and the process information on the individual code do not conflict with each other, other settings included in the process information of each code may be set as the setting of the information processing apparatus **10**.
(73) For example, in the process information, the storage destination and the setting of the OCR process do not conflict with each other. Therefore, even in a case where the individual code including the OCR process setting as the process information is scanned after the face sheet code including the storage destination as the process information is scanned, the information processing apparatus **10** may apply the setting of the OCR process related to the process information extracted from the individual code to execute the process. In addition, by providing a priority order for each setting, and by applying the setting related to process information extracted from an individual code according to the priority order, whether or not a process related to other settings is executed

may be determined and the process may be executed.

Second Exemplary Embodiment

(74) In the first exemplary embodiment, the mode of executing a process of storing the corresponding image data according to the process information included in the extracted code is described. In the present exemplary embodiment, a mode of presenting to a user a process to be executed according to the process information included in the extracted code will be described.

(75) A configuration of an information processing system (see FIG. **1**), a process related to a face sheet code (see FIG. **2**), a process related to an individual code (see FIG. **3**), and a hardware configuration of the information processing apparatus **10** (see FIG. **4**) according to the present exemplary embodiment have the same manner as the first exemplary embodiment, so the description thereof will be omitted. Further, a schematic diagram (see FIGS. **6** to **10**) illustrating a process of capturing a plurality of documents and a flowchart illustrating a flow of a setting process (see FIG. **12**) according to the present exemplary embodiment have the same manner as the first exemplary embodiment, so the description thereof will be omitted.

(76) A functional configuration of the information processing apparatus **10** will be described with reference to FIG. **13**. FIG. **13** is a block diagram illustrating an example of a functional configuration of the information processing apparatus **10** according to the present exemplary embodiment. In FIG. **13**, the identical functions as the functional configuration illustrated in FIG. **4** are denoted by the identical reference numerals with the reference numerals in FIG. **4**, and the description thereof will be omitted.

(77) As an example, as illustrated in FIG. **13**, the information processing apparatus **10** includes the acquisition unit **21**, the extraction unit **22**, the setting unit **23**, and the processing unit **24**B. In a case where the CPU **11** executes the information processing program to function as the acquisition unit **21**, the extraction unit **22**, the setting unit **23**, and the processing unit **24**B.

(78) The processing unit **24**B associates target image data with a set storage destination by using a setting in the information processing apparatus **10**, and presents the associated image data and storage destination to the user. The processing unit **24**B performs a process of storing the associated image data in the storage destination according to an instruction of the user.

(79) In addition, the processing unit **24**B accepts selection of the user from the presented image data, and performs a process of displaying a preview of the selected image data.

(80) As an example, as illustrated in FIG. **14**, the processing unit **24**B displays a captured document display screen **40** to present document information on associated image data and storage destination to the user. Here, the captured document display screen **40** includes a captured result display region **41**, an execute button **42**, and a cancel button **43**. The captured result display region **41** is a region for displaying the document information on the associated image data and storage destination according to the extracted code by capturing a document. Further, the captured result display region **41** includes a target display region **45** that is classified and displayed for each document to be processed. Since the target display region **45** classifies the document for each process to be executed, it is possible to easily recognize which process is executed for the document and which process is executed to be included in the process.

(81) The execute button **42** accepts an instruction to execute the subsequent process by being pressed by the user. The subsequent process is a process of storing the image data in the storage destination, according to the associated image data and storage destination. The cancel button **43** accepts an instruction not to execute the subsequent process by being pressed by the user.

(82) As illustrated in FIG. **14**, in a case where a document including a plurality of codes is captured, the captured document display screen **40** displays a target, a storage destination, and other settings related to each document in the captured result display region **41**. In a case where a document including the face sheet code **30** and the individual code **34** is captured, and processes conflict with each other and image data is stored in a storage destination different from the storage destination included in the code, the captured document display screen **40** displays that the process

corresponding to the code is not executed. For example, the captured document display screen **40** displays a text string **44** indicating that a process corresponding to a code is not executed on a target document is stored in a storage destination different from the storage destination included in the code, in the captured result display region **41**.

(83) Further, in the present exemplary embodiment, regarding the process displayed on the captured document display screen **40** illustrated in FIG. **14**, the mode in which the process corresponding to a face sheet code of a face sheet document **1** is executed on an individual document **2**, so that the process corresponding to an individual code of the individual document **2** is not executed is described. Meanwhile, the exemplary embodiment is not limited thereto. In a case where the process corresponding to the individual code cannot be executed as a result of executing the process corresponding to the face sheet code, as a process different from the process corresponding to the individual code, an alternative process of the process corresponding to the individual code may be proposed and controlled to be executed. For example, as the process corresponding to the individual code, a shortcut to the image data corresponding to the individual document **2** stored in the folder B or stored in the folder A is stored in the folder B. Further, in a case where the other setting is that a process corresponding to the face sheet code is a process of converting the document to a predetermined file format A and a process corresponding to the individual code is a process of converting the document to a predetermined file format B, and it is determined that these processes conflict with each other, only the process of converting to the predetermined file format A, which is the process corresponding to the face sheet code, is executed. In such a case, as a different process of substituting the process corresponding to the individual code, a process of duplicating image data corresponding to the document and converting the image data into the predetermined file format B may be executed.

(84) The captured document display screen **40** accepts selection of the document information displayed in the captured result display region **41**, and displays a preview of the image data related to the selected document information.

(85) Next, an action of the information processing apparatus **10** according to the present exemplary embodiment will be described with reference to FIG. **15**. FIG. **15** is a flowchart illustrating an example of a document capturing process according to the present exemplary embodiment. In a case where the CPU **11** reads an information processing program from the ROM **12** or the storage **14**, and executes the information processing program to execute a document capturing process illustrated in FIG. **15**. The document capturing process illustrated in FIG. **15** is executed, for example, in a case where an instruction to execute a process and image data are input from the multi-function device **2**. In FIG. **15**, the identical steps as the document capturing process illustrated in FIG. **11** are denoted by the identical reference numerals with the reference numerals in FIG. **11**, and description thereof will not be repeated.

(86) In step S**107**, the CPU **11** determines whether or not the face sheet code **30** or a code indicating the individual code **34** is scanned from the scanning page. In a case where the code is scanned (YES in step S**107**), the CPU **11** proceeds to step S**103**. On the other hand, in a case where the code cannot be scanned from the scanning page (NO in step S**107**), the CPU **11** proceeds to step S**108**.

(87) In step S**108**, the CPU **11** uses the target and the storage destination set in the information processing apparatus **10** to store the image data corresponding to the scanning page and the storage destination in association with each other. Here, in a case where the individual code **34** is extracted after the face sheet code **30** is extracted, the text string **44** indicating that the process corresponding to the code is not executed is further associated and stored, for the page **35** (document) to which the individual code **34** is added.

(88) In step S**109**, the CPU **11** displays the captured document display screen **40**, and presents the captured result.

(89) In step S**110**, the CPU **11** determines whether or not the execute button **42** is pressed on the

captured document display screen **40**. In a case where the execute button **42** is pressed (YES in step S**110**), the CPU **11** proceeds to step S**111**. On the other hand, in a case where the execute button **42** is not pressed (NO in step S**110**), the CPU **11** proceeds to step S**112**.
(90) In step S**111**, the CPU **11** stores the image data in the associated storage destination.
(91) In step S**112**, the CPU **11** determines whether or not document information is selected on the captured document display screen **40**. In a case where the document information is selected (YES in step S**112**), the CPU **11** proceeds to step S**113**. On the other hand, in a case where the document information is not selected (the cancel button **43** is pressed) (NO in step S**112**), the CPU **11** ends the process.
(92) In step S**113**, the CPU **11** displays a preview of the image data related to the selected document information.
(93) As described above, according to the present exemplary embodiment, in a case where a page to which a first code is added and a page to which a second code is added are mixed, the user can recognize that the process represented by process information included in the first code and the process represented by process information included in the second code conflict with each other.
Modification Example of Second Exemplary Embodiment
(94) The captured result display region **41** according to the present exemplary embodiment is described as the mode in which the documents are classified and displayed for each document. Meanwhile, the exemplary embodiment is not limited thereto. The documents may be classified and displayed for each process to be executed. For example, as illustrated in FIG. **16**, the captured result display region **41** may display the image data corresponding to the process target, the storage destination, the other settings, and the captured page applied for each process represented by the extracted process information. In addition, the captured result display region **41** may display a thumbnail **46** corresponding to the image data on which the process is executed. Further, in a case where a process different from the process included in the code is executed on the image data, the thumbnail **46** related to the image data may be highlighted, in the captured result display region **41**. Specifically, in FIG. **16**, in a case where a document to which a “process by face sheet **1**” is associated includes a document to which an individual code is added, image data related to the document to which the individual code is added may be highlighted and displayed as a thumbnail **46**A.
(95) Although the exemplary embodiment of the invention is described by using each of the exemplary embodiments, the exemplary embodiment of the invention is not limited to the scope described in each of the exemplary embodiments. Various modifications or improvements can be added to each of the exemplary embodiments without departing from the gist of the present exemplary embodiment of the invention and the modified or improved form is also included in the technical scope of the present exemplary embodiment of the invention.
(96) In the present exemplary embodiment, the mode of accepting an instruction as to whether or not to execute a subsequent process is described. Meanwhile, the exemplary embodiment is not limited thereto. An instruction as to whether or not to capture each document as image data may be accepted.
(97) In the embodiments above, the term “processor” refers to hardware in a broad sense. Examples of the processor include general processors (e.g., CPU: Central Processing Unit) and dedicated processors (e.g., GPU: Graphics Processing Unit, ASIC: Application Specific Integrated Circuit, FPGA: Field Programmable Gate Array, and programmable logic device).
(98) In the embodiments above, the term “processor” is broad enough to encompass one processor or plural processors in collaboration which are located physically apart from each other but may work cooperatively. The order of operations of the processor is not limited to one described in the embodiments above, and may be changed.
(99) Further, in the present exemplary embodiment, the mode in which the information processing program is installed in the storage is described, and the present exemplary embodiment is not

limited thereto. The information processing program according to the present exemplary embodiment also may be provided as a mode of being recorded in a computer readable storage medium. For example, the information processing program according to the exemplary embodiment of the present invention may be provided in a mode of being recorded on an optical disc such as a compact disc (CD)-ROM and a digital versatile disc (DVD)-ROM. The information processing program according to the exemplary embodiment of the present invention may be provided in a mode of being recorded in a semiconductor memory such as a Universal Serial Bus (USB) memory and a memory card. Further, the information processing program according to the present exemplary embodiment may be acquired from an external apparatus via a communication line connected to a communication I/F.

(100) The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

## Claims

1. An information processing apparatus comprising: a processor configured to: in a case where a first page to which a first code including process information, which represents information for processing a document and in which at least another page other than the first page is set as a process target, is added and a second page to which a second code including process information, which represents information for processing a document and in which the second page is set as the process target, is added are captured, execute a process represented by the process information of the first code without executing a process represented by the process information of the second code in a case where the process represented by the process information of the first code conflicts with the process represented by the process information of the second code.
2. The information processing apparatus according to claim 1, wherein the processor is configured to: in a case where the process represented by the process information of the first code does not conflict with the process represented by the process information of the second code, execute each process represented by the process information on each corresponding process target.
3. The information processing apparatus according to claim 1, wherein the processor is configured to: in a case where the second code is scanned before the first code is scanned, execute the process represented by the process information of the second code.
4. The information processing apparatus according to claim 2, wherein the processor is configured to: in a case where the second code is scanned before the first code is scanned, execute the process represented by the process information of the second code.
5. The information processing apparatus according to claim 3, wherein the processor is configured to: until the first code is scanned, scan the second code added to each page, and execute each process represented by the process information of the second code.
6. The information processing apparatus according to claim 4, wherein the processor is configured to: until the first code is scanned, scan the second code added to each page, and execute each process represented by the process information of the second code.
7. The information processing apparatus according to claim 1, wherein the processor is configured to: in a case where another first code is scanned after the first code is scanned, set a page from a page after the first page to a page before another first page corresponding to the other first code as the process target of the process represented by the process information of the first code.

8. The information processing apparatus according to claim 2, wherein the processor is configured to: in a case where another first code is scanned after the first code is scanned, set a page from a page after the first page to a page before another first page corresponding to the other first code as the process target of the process represented by the process information of the first code.
9. The information processing apparatus according to claim 3, wherein the processor is configured to: in a case where another first code is scanned after the first code is scanned, set a page from a page after the first page to a page before another first page corresponding to the other first code as the process target of the process represented by the process information of the first code.
10. The information processing apparatus according to claim 4, wherein the processor is configured to: in a case where another first code is scanned after the first code is scanned, set a page from a page after the first page to a page before another first page corresponding to the other first code as the process target of the process represented by the process information of the first code.
11. The information processing apparatus according to claim 7, wherein the processor is configured to: in a case where the other first code is scanned after the first code is scanned, set a page after the other first page as a process target of a process represented by process information of the other first code.
12. The information processing apparatus according to claim 1, wherein the processor is configured to: execute a process of storing image data corresponding to a page, which is a process target.
13. The information processing apparatus according to claim 12, wherein the process information further includes storage destination information which is information representing a storage destination for storing the image data, and the processor is configured to: store image data corresponding to the page, which is the process target, according to the storage destination information represented by the process information.
14. The information processing apparatus according to claim 1, wherein the processor is configured to: associate process information indicating a process to be executed with the captured document, and display the document and the process information associated with the document.
15. The information processing apparatus according to claim 14, wherein the processor is configured to: in a case where the process information of the first code is associated with the document including the second page, notify that the process represented by the process information of the first code is executed without executing the process represented by the process information of the second code related to the document including the second page.
16. The information processing apparatus according to claim 15, wherein the processor is configured to: execute an alternative process different from the process represented by the process information of the second code in addition to the process represented by the process information of the first code, on the document including the second page.
17. The information processing apparatus according to claim 14, wherein the processor is configured to: in a case where the process information of the first code is associated with the document including the second page, accept whether or not to execute a process different from the process represented by the process information of the second code related to the document.
18. The information processing apparatus according to claim 14, wherein the processor is configured to: in a case where the process information of the first code is associated with the document including the second page, display a page, which is the second page and on which the process represented by the process information of the second code is not executed, to be distinguished from other pages.
19. A non-transitory computer readable medium storing an information processing program causing a computer to execute a process comprising: executing, in a case where a first page to which a first code including process information, which represents information for processing a document and in which at least another page other than the first page is set as a process target, is added and a second page to which a second code including process information, which represents information for processing a document and in which the second page is set as the process target, is added are

captured, a process represented by the process information of the first code without executing a process represented by the process information of the second code in a case where the process represented by the process information of the first code conflicts with the process represented by the process information of the second code.

20. An information processing method comprising: executing, in a case where a first page to which a first code including process information, which represents information for processing a document and in which at least another page other than the first page is set as a process target, is added and a second page to which a second code including process information, which represents information for processing a document and in which the second page is set as the process target, is added are captured, a process represented by the process information of the first code without executing a process represented by the process information of the second code in a case where the process represented by the process information of the first code conflicts with the process represented by the process information of the second code.